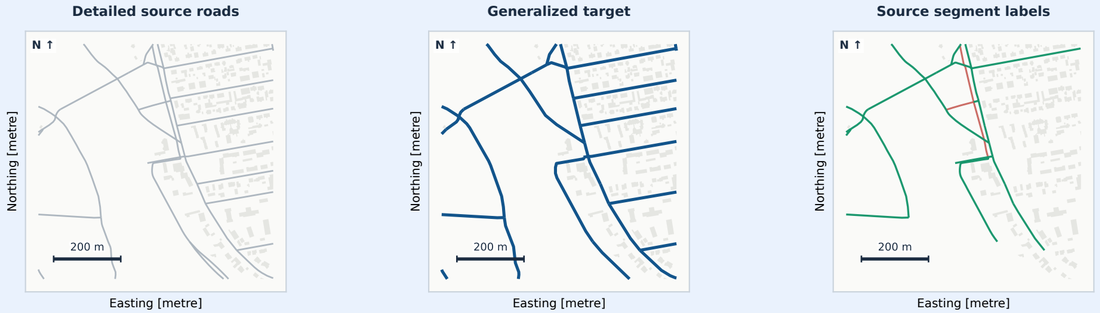
CartoGraph-X | Context-Aware Road Selection for Cartographic Generalization

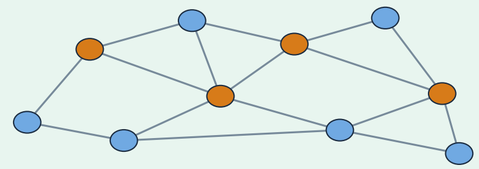
Real georeferenced road scenes: Vilnius · Proximity graph · Dual-feature encoding · Country-conditioned prediction

1. Input data and road correspondence



Geometric matching: search 15 m · buffer 5 m · source ≥ 0.20 · target ≥ 0.03

2. Road graph construction



kNN proximity, k = 6; distance ≤ 250 m

Edges do not assert physical junctions

3. Feature extraction

Road-level geometry

- Length (m)
- Intersections
- Road count (150 m)
- Mean neighbor distance

Graph context

- Spatial neighborhood
- Graph descriptors
- Local road structure
- Contextual statistics

4. CartoGraph-X model architecture

Two complementary feature branches

Road features
4 geometric

Graph context
19 features

Road feature encoder
MLP

Residual GraphSAGE
2 message-passing layers

Gated feature fusion
z = g ⊙ h_graph + (1−g) ⊙ h_road

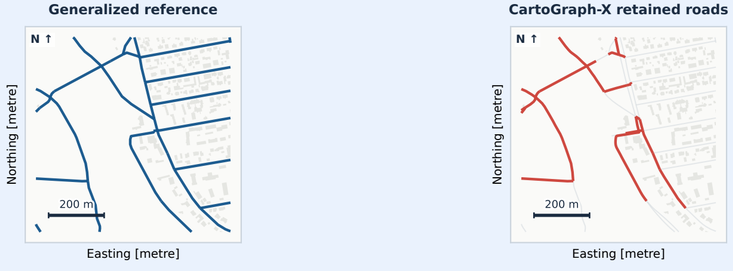
Shared classifier
MLP + sigmoid

Lithuania head
joint training

Poland head
joint training

Binary retention probability $p_i = P(y_i = 1 \mid \text{road } i, \text{spatial context})$

5. Real road-selection output



Source geometries

selected by the model

Not geometric simplification

6. Experimental evaluation and efficiency

Six evaluation protocols
LT, PL, zero-shot, joint

Predictive performance
MCC · BA · ROC/PR-AUC

Computational efficiency
10 seeds · time · GPU · FLOPs

GIS maps are rendered from Script 124 georeferenced scene layers; no synthetic roads or invented predictions.

C2 matching labels provisional · Geographic test findings exploratory